

Assignment - 1

Question 1

Write a SQL statement to create a simple table countries including columns country_id, country_name and region_id.

```
CREATE TABLE countries (  
    country_id VARCHAR(2),  
    country_name VARCHAR(40),  
    region_id DECIMAL(10,0)  
);
```

Output :-

```
mysql> use db  
Database changed  
mysql> CREATE TABLE countries (  
    ->     country_id VARCHAR(2),  
    ->     country_name VARCHAR(40),  
    ->     region_id DECIMAL(10,0)  
    -> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> select * from countries;  
Empty set (0.00 sec)
```

Question 2

Write a SQL statement to create a simple table countries including columns country_id, country_name and region_id which is already exists.

```
CREATE TABLE IF NOT EXISTS countries (  
    country_id VARCHAR(2),  
    country_name VARCHAR(40),  
    region_id DECIMAL(10,0)  
);
```

Output :-

```
mysql> CREATE TABLE IF NOT EXISTS countries (  
->     country_id VARCHAR(2),  
->     country_name VARCHAR(40),  
->     region_id DECIMAL(10,0)  
-> );  
Query OK, 0 rows affected, 1 warning (0.00 sec)  
  
mysql> select * from countries ;  
Empty set (0.00 sec)
```

Question 3

Write a SQL statement to create the structure of a table dup_countries similar to countries.

CREATE TABLE dup_countries LIKE countries;

```
mysql> CREATE TABLE dup_countries LIKE countries;  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> select * from dup_countries;  
Empty set (0.00 sec)
```

Question 4

Write a SQL statement to create a duplicate copy of countries table including structure and data by name dup_countries.

```
CREATE TABLE dup_countries AS SELECT * FROM countries;
```

Output :-

```
mysql> CREATE TABLE dup_countries AS SELECT * FROM countries;
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from dup_countries;
Empty set (0.00 sec)
```

Question 5

Write a SQL statement to create a table countries set a constraint NULL.

```
CREATE TABLE countries (  
    country_id VARCHAR(2) NOT NULL,  
    country_name VARCHAR(40) NOT NULL,  
    region_id DECIMAL(10,0) NOT NULL  
);
```

Output :-

```
mysql> CREATE TABLE countries (  
->     country_id VARCHAR(2) NOT NULL,  
->     country_name VARCHAR(40) NOT NULL,  
->     region_id DECIMAL(10,0) NOT NULL  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

Question 6

Write a SQL statement to create a table named jobs including columns job_id, job_title, min_salary, max_salary and check whether the max_salary amount exceeding the upper limit 25000.

```
CREATE TABLE jobs (  
    job_id VARCHAR(10) NOT NULL,  
    job_title VARCHAR(35) NOT NULL,  
    min_salary DECIMAL(6,0),  
    max_salary DECIMAL(6,0),  
    CHECK (max_salary <= 25000)  
);
```

Output :-

```
mysql> CREATE TABLE jobs (  
->     job_id VARCHAR(10) NOT NULL,  
->     job_title VARCHAR(35) NOT NULL,  
->     min_salary DECIMAL(6,0),  
->     max_salary DECIMAL(6,0),  
->     CHECK (max_salary <= 25000)  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> select * from jobs ;  
Empty set (0.00 sec)
```

Question 7

Write a SQL statement to create a table named countries including columns country_id, country_name and region_id and make sure that no countries except Italy, India and China will be entered in the table.

```
CREATE TABLE countries (  
    country_id VARCHAR(2),  
    country_name VARCHAR(40),  
    region_id DECIMAL(10,0),  
    CHECK (country_name IN ('Italy', 'India', 'China'))  
);
```

Output :-

```
mysql> CREATE TABLE countries (  
->     country_id VARCHAR(2),  
->     country_name VARCHAR(40),  
->     region_id DECIMAL(10,0),  
->     CHECK (country_name IN ('Italy', 'India', 'China'))  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql>  
mysql> select * from countries;  
Empty set (0.00 sec)
```

Question 8

Write a SQL statement to create a table named job_history including columns employee_id, start_date, end_date, job_id and department_id and make sure that the value against column end_date will be entered at the time of insertion to the format like '--/--/----'.

```
CREATE TABLE job_history (  
    employee_id DECIMAL(6,0) NOT NULL,  
    start_date DATE NOT NULL,  
    end_date DATE NOT NULL,  
    job_id VARCHAR(10) NOT NULL,  
    department_id DECIMAL(4,0)  
);
```

Output :-

```
mysql> CREATE TABLE job_history (  
->     employee_id DECIMAL(6,0) NOT NULL,  
->     start_date DATE NOT NULL,  
->     end_date DATE NOT NULL,  
->     job_id VARCHAR(10) NOT NULL,  
->     department_id DECIMAL(4,0)  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql>  
mysql> select * from job_history;  
Empty set (0.00 sec)
```

Question 9

Write a SQL statement to create a table named countries including columns country_id, country_name and region_id and make sure that no duplicate data against column country_id will be allowed at the time of insertion.

```
CREATE TABLE countries (  
    country_id VARCHAR(2),  
    country_name VARCHAR(40),  
    region_id DECIMAL(10,0),  
    UNIQUE (country_id)  
);
```

Output :-

```
mysql> CREATE TABLE countries (  
->     country_id VARCHAR(2),  
->     country_name VARCHAR(40),  
->     region_id DECIMAL(10,0),  
->     UNIQUE (country_id)  
-> );  
Query OK, 0 rows affected (0.06 sec)  
  
mysql>  
mysql> select * from countries ;  
Empty set (0.00 sec)
```

Question 10

Write a SQL statement to create a table named jobs including columns job_id, job_title, min_salary and max_salary, and make sure that, the default value for job_title is blank and min_salary is 8000 and max_salary is NULL will be entered automatically at the time of insertion if no value assigned for the specified columns.

```
CREATE TABLE jobs (  
    job_id VARCHAR(10) NOT NULL,  
    job_title VARCHAR(35) NOT NULL DEFAULT '',  
    min_salary DECIMAL(6,0) DEFAULT 8000,  
    max_salary DECIMAL(6,0) DEFAULT NULL  
);
```

Output :-

```
mysql> CREATE TABLE jobs (  
->     job_id VARCHAR(10) NOT NULL,  
->     job_title VARCHAR(35) NOT NULL DEFAULT ' ',  
->     min_salary DECIMAL(6,0) DEFAULT 8000,  
->     max_salary DECIMAL(6,0) DEFAULT NULL  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql>  
mysql> select * from jobs;  
Empty set (0.00 sec)
```

Question 11

Write a SQL statement to create a table named countries including columns country_id, country_name and region_id and make sure that the country_id column will be a key field which will not contain any duplicate data at the time of insertion.

```
CREATE TABLE countries (  
    country_id VARCHAR(2),  
    country_name VARCHAR(40),  
    region_id DECIMAL(10,0),  
    PRIMARY KEY (country_id)  
);
```

Output :-

```
mysql> CREATE TABLE countries (  
->     country_id VARCHAR(2),  
->     country_name VARCHAR(40),  
->     region_id DECIMAL(10,0),  
->     PRIMARY KEY (country_id)  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> select * from countries ;  
Empty set (0.00 sec)
```

Question 12

Write a SQL statement to create a table countries including columns country_id, country_name and region_id and make sure that the column country_id will be unique and store an auto incremented value.

```
CREATE TABLE countries (  
    country_id INT NOT NULL AUTO_INCREMENT,  
    country_name VARCHAR(40),  
    region_id DECIMAL(10,0),  
    PRIMARY KEY (country_id)  
);
```

Output :-

```
mysql> CREATE TABLE countries (  
->     country_id INT NOT NULL AUTO_INCREMENT,  
->     country_name VARCHAR(40),  
->     region_id DECIMAL(10,0),  
->     PRIMARY KEY (country_id)  
-> );  
Query OK, 0 rows affected (0.06 sec)  
  
mysql>  
mysql> select * from countries;  
Empty set (0.00 sec)
```

Question 13

Write a SQL statement to create a table countries including columns country_id, country_name and region_id and make sure that the combination of columns country_id and region_id will be unique.

```
CREATE TABLE countries (  
    country_id VARCHAR(2),  
    country_name VARCHAR(40),  
    region_id DECIMAL(10,0),  
    UNIQUE (country_id, region_id)  
);
```

Output :-

```
mysql> CREATE TABLE countries (  
->     country_id VARCHAR(2),  
->     country_name VARCHAR(40),  
->     region_id DECIMAL(10,0),  
->     UNIQUE (country_id, region_id)  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql>  
mysql>  
mysql> select * from countries;  
Empty set (0.00 sec)
```